

UBERLANDIA, Brazil

WMO: 835250

Lat: 18.883S

Lon: 48.220W

Elev: 3094

StdP: 13.13

Time Zone: -3.00 (SAB)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	51.8	55.3	33.6	31.6	74.0	37.1	36.3	75.1	19.0	70.6	17.7	70.1	10.8	100	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	18.8	90.0	63.8	88.1	64.8	86.4	65.8	73.2	80.2	72.3	79.8	71.6	79.4	8.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.6	131.1	75.8	70.2	124.7	74.4	69.8	123.1	74.0	39.1	79.7	38.2	78.8	37.6	79.5	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	17.0	15.3	DB	44.0	93.6	2.8	2.5	42.0	95.4	40.3	96.9	38.7	98.3	36.6	100.1	(4)
(5)			WB	41.2	73.7	3.7	1.7	38.5	74.9	36.4	75.9	34.3	76.9	31.7	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.6	74.2	75.0	74.2	72.8	69.3	68.0	67.9	70.8	74.7	75.6	74.1	74.0		
	DBStd	4.24	2.74	2.27	2.35	2.89	3.23	3.10	3.91	3.70	4.50	4.34	2.76	2.60		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	45	0	0	0	0	7	10	20	5	2	1	0	0		
	CDD50	8232	751	701	751	685	598	541	555	646	740	795	724	745		
	CDD65	2801	286	281	286	235	140	101	109	186	292	331	274	280		
	CDH74	17626	1576	1665	1497	1239	667	488	680	1421	2597	2705	1607	1484		
	CDH80	4543	385	422	320	210	49	20	62	318	990	1037	402	328		
(14) Wind	WSAvg	8.3	7.8	7.4	7.3	7.6	7.9	8.3	8.8	9.4	9.7	9.2	8.5	7.8		
(15) Precipitation	PrecAvg	55.1	11.0	8.1	8.2	2.6	1.5	0.5	0.2	0.6	1.9	3.7	7.1	9.9		
	PrecMax	64.4	18.6	14.4	14.2	5.1	4.3	3.8	1.0	1.8	4.0	7.1	10.5	14.9		
	PrecMin	46.7	4.6	1.7	3.5	0.7	0.0	0.0	0.0	0.0	0.0	1.7	4.0	6.5		
	PrecStd	5.5	3.9	3.8	3.1	1.2	1.0	0.9	0.3	0.6	1.2	1.5	1.9	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.0	88.2	86.4	86.4	82.8	82.2	84.0	88.0	93.1	94.8	89.2	87.6		
		MCWB	67.5	69.5	70.0	69.6	65.0	62.6	60.4	60.7	62.3	63.8	67.8	68.5		
	2%	DB	86.0	86.2	85.6	84.3	80.9	80.4	82.1	85.8	90.0	91.2	86.2	85.8		
		MCWB	68.3	69.2	70.0	68.7	64.5	62.6	60.6	60.1	62.3	64.2	67.9	68.8		
	5%	DB	84.2	84.5	83.9	82.5	79.2	78.7	80.4	83.8	88.1	88.2	84.3	83.9		
		MCWB	68.9	69.1	69.7	68.3	64.4	62.3	60.2	60.2	61.9	64.8	68.3	69.0		
	10%	DB	82.3	82.6	82.1	80.7	78.4	77.0	78.5	81.0	86.0	86.0	82.4	82.0		
		MCWB	69.3	69.5	69.5	67.8	64.2	62.0	60.1	59.6	61.8	65.1	68.5	69.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.7	74.5	73.6	71.9	69.3	67.5	66.3	66.6	70.7	71.8	73.4	74.4		
		MCDB	79.9	81.6	81.8	81.5	77.4	76.3	76.0	76.6	79.8	81.6	80.4	79.3		
	2%	WB	72.5	72.9	72.3	71.0	68.2	66.0	64.0	64.2	68.7	70.5	71.8	72.8		
		MCDB	79.2	80.3	80.6	80.0	76.4	75.5	75.3	76.0	76.9	79.8	79.8	78.8		
	5%	WB	71.5	71.9	71.5	70.2	66.9	64.9	62.9	62.9	67.1	69.3	70.9	71.7		
		MCDB	78.6	79.5	79.7	78.5	74.8	74.5	74.3	75.8	76.3	78.4	78.8	78.3		
	10%	WB	70.7	70.9	70.8	69.3	65.7	63.7	61.7	61.7	65.7	68.5	69.9	70.8		
		MCDB	77.4	78.4	78.4	76.9	73.7	73.0	73.7	75.3	76.4	77.8	77.6	77.4		
(35) Mean Daily Temperature Range	MDBR		15.1	16.0	14.9	16.4	17.6	18.7	19.8	21.2	20.8	18.8	16.3	15.1		
	5% DB	MCDBR	18.2	18.3	17.5	17.8	18.6	19.4	21.1	22.3	22.6	21.5	19.1	18.1		
		MCWBR	6.3	6.4	6.4	6.6	7.3	7.5	7.7	7.6	7.3	6.6	6.4	6.4		
	5% WB	MCDBR	14.9	16.0	15.5	15.8	16.0	17.3	19.2	20.3	18.8	17.9	16.6	15.0		
		MCWBR	6.6	6.7	6.8	6.4	7.0	7.7	8.0	7.9	7.3	6.9	6.6	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.374	0.374	0.363	0.350	0.329	0.321	0.333	0.360	0.448	0.426	0.391	0.382		
	taud		2.496	2.481	2.529	2.532	2.550	2.578	2.507	2.408	2.172	2.291	2.412	2.460		
	Ebn at Noon		308	305	300	292	287	284	283	285	270	285	300	305		
	Edh at Noon		37	37	34	31	29	27	30	35	48	44	40	38		
(44) All-Sky Solar Radiation	RadAvg		1816	1891	1717	1657	1474	1395	1477	1721	1808	1876	1775	1814		
	RadStd		214	157	148	107	67	57	71	87	153	153	102	133		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page